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Project Title:	Mechanisms of the Repair of HZE Induced DNA Double-Strand Breaks in Human Cells		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
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Space Biology Special Category:	None		
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Comments:			
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No. of Master's Candidates:		**No. of Bachelor's Degrees:**	0
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Flight Program:			
Flight Assignment:	NOTE: Project ended early and new grant awarded for new direction to this research, per A. Chu/ARC. See project with same title and grant #NNX12AH07G (Ed., 5/24/12) NOTE: End date is now 9/30/2012 per A. Chu/NSSC. Also, change in Gaps per HRR information (Ed., 9/26/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Asaithamby, Aroumougame (University of Texas Southwestern Medical Center at Dallas)		
Grant/Contract No.:	NNX07AU42G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

[EDITOR'S NOTE 5/14/2012: Project ended early and new grant awarded for new direction to this research, per A. Chu/ARC. See project with same title and grant #NNX12AH07G for future reports]
Overall Objective

DNA repair capability is one of the key indicators reflecting the biological consequences of cellular exposure to genotoxic insults. Determining the biochemical nature of the DNA damages induced by HZE particles and identifying the factors that are required for the repairing of these DNA lesions will provide tools to assess individual radiation susceptibility and also to validate risk assessment for human exposure to HZE particles. The goals of this proposal are two fold: 1) We will develop new approaches to determine the types and spatial relationship among different DNA lesions formed along the dense ionizing tracks induced by HZE particles in human cells and to dissect the cellular capability to repair these DNA lesions in situ; and 2) we will identify molecular pathways or factors that may be able to help the human cell respond efficiently to HZE particle-induced DNA lesions. For this purpose, we will test the hypothesis that specific nucleases, Artemis and/or Werner Syndrome protein (WRN), contribute specifically to the repair of complex DNA lesions induced by HZE particles. We will also determine the role of homologous recombination factors in repair of HZE particle-induced DSBs (double strand breaks). With these new approaches on hand, we should be able to provide a semi quantitative measurement for the generation and spatial distribution of clustered DNA lesions along the dense track induced by HZE particles and determine the relative contribution of these Nonhomologous end-joining and homologous end joining factors in the repair of DNA DSBs, but also able to identify whether these repair activities play any role in managing clustered DNA lesions generated by HZE. These studies would, in turn, provide biological markers for risk assessment and guidance in radiation protection for astronauts during space missions in the future.

Specific Objectives

A.1. To verify that multiple types of DNA lesions formed in close vicinity along the dense ionizing tracks induced by HZE particles in situ. We will use antibodies for repair enzymes, fluorescent tagged repair proteins as surrogate markers as well as direct assay system to identify various types of DNA lesions and also monitor the repair status of the DNA damage in human cells exposed to HZE particles.

1) To verify whether DNA lesions including base damage, DNA single strand breaks (SSB), and DSBs are clustered along the tracks induced by HZE particles.

2) To determine the repair of clustered DNA lesions formed along the tracks induced by HZE particles.

3) To determine the nature of the persistent or “unrepaired” DNA lesions induced by HZE and its spatial relationship to clustered DNA damages.

A.2. To test the hypothesis that Artemis, a specific exonuclease/endonuclease, which may be involved in the repair of complex DNA DSBs induced by HZE particle through nonhomologous end joining (NHEJ) pathway. Evidence suggest that Artemis protein, a specific 5’ to 3’ exonuclease/endonuclease, which play a key role in NHEJ, but specially play a role in the repair of complex DNA DSBs induced by high LET radiation. We will determine the functions of Artemis in the repair of DSBs induced by HZE particles.

1) To determine the radiosensitivity and repair of DNA DSBs and clustered DNA lesions induced by various HZE particles in human Artemis deficient cell lines.

2) To determine whether the Artemis’ exonuclease/endonuclease activity is involved in the repair of HZE induced DSBs. We will measure the radiosensitivity and repair of DSBs induced by HZE in human Artemis nuclease dead cell line.

3) To determine the role of Artemis in recognition of HZE particle induced DSBs. We will use green fluorescent protein (GFP)-tagged Artemis to measure the initial and kinetics of the Artemis localization on DNA damage sites induced by various HZE particles.

A.3. To test the hypothesis that WRN, a human multifunctional protein with both exonuclease and helicase activities, protein may involve in managing the repair of HZE induced complex subset of DNA DSBs. Our preliminary data suggests that human Werner syndrome patient cells exposed to HZE particles result in elevated unrepaired DSBs as compared to those induced by gamma-rays. However, the role of WRN in DSB repair, especially in HZE induced DSBs, is not clear. We plan to determine the role of WRN in processing complex DSBs in HZE irradiated cells.

1) To determine the radiosensitivity and repair of DNA DSBs and clustered DNA lesions induced by various HZE particles in human WRN deficient cell lines.

2) To determine the roles of exonuclease and helicase activities of WRN in repair of HZE particles induced DNA DSBs and clustered DNA lesions.

3) To determine whether WRN localization on DNA damage sites is dependent on the quality of HZE particles and also whether WRN phosphorylation is required for the repair of HZE induced DSBs.

A.4. To determine the role of homologous recombination (HR) in the repair of complex DSBs induced by HZE particles. Indirect evidence suggests that HR frequency might be increased in cells exposed to high LET radiation. However, the role of HR in the repair of HZE particle induced DSBs in human cells is completely unknown. We will use unique human knockout mutant cell lines to address this issue.

1) To determine whether eliminating both HR and NHEJ pathways would elevate unrepaired DNA DSBs and enhance cellular sensitivity to HZE particle irradiation. We have constructed a set of isogenic knockout human cell lines that are deficient in HR (Rad54-/-), NHEJ (ligase IV-/-) and HR/NHEJ (Rad54-/- and Ligase IV-/-), and will determine the radiation sensitivity and DSB repair of these cell lines in response to HZE particles.

2) To determine whether the function of WRN in response to HZE particle is involved in HR or NHEJ of the DNA DSB repair pathway. In order to answer this question, we plan to measure whether there is a synergistic radiobiological effect in WRN-/-/Rad54-/- or WRN-/-/ligase IV-/- double knockout human cell line as compared to individual knock out cell lines.

EDITOR's NOTE (5/24/2012): Project ended early and new grant awarded for new direction to this research, per A. Chu/ARC. See "Mechanisms of the Repair of HZE Induced DNA Double-Strand Breaks in Human

	Cells--NNX12AH07G" for future reports.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research would lead to the development of counter-measures to enhance astronauts' ability to sustain HZE particles and radiation exposures in space.
Task Progress:	SUPPLEMENTAL INFORMATION FOR FINAL REPORTING (AUGUST 2012) Overall, we have established that 1. HR-deficient hamster cells are highly sensitive to iron particles as compared with wild-type cells. 2. As DSBs induced by HZE particles are complex, and WRN has shown to have roles in NHEJ, HR, and BER pathways, it might play one or more roles in HZE particle induced DNA repair either by coordinating within the individual DNA repair pathways or it may coordinate between them. This could be achieved either by binding directly to the damaged DNA or recruited by other DNA-repair factors. 3. Artemis might be needed to process damaged ends at DSBs prior to rejoining. This might require DNA-PK- and ATM-dependent phosphorylation of Artemis followed by its activity modification from exonuclease to endonuclease. 4. FANCA-deficient human cells are highly sensitive to Fe particles as compared with FANCA deficient cells complemented with WT FANCA, suggesting a role for FA complex in the repair of clustered DNA lesions. However, whether all the factors of the FA complex orchestrate the repair of complex DNA lesions and the method by which these factors function in this processes is totally unknown. REPORTING FOR ANNUAL PROGRESS FROM 08/01/2010-07/31/2011: Overall, we have established that 1. Using this in situ approach we confirmed that most DNA damage induced by Fe and Si ions is complex in nature and that the yield and the complexity of damage increased with an increase in the LET of the radiation. 2. A tight spatial distribution of different types of lesions within the clustered DNA damage induced by Fe ions resulted in lesions that were slowly repaired or irreparable. 3. Difficulties associated with complex DNA lesions repair were not due to their physical location within the sub-nuclear domains. 4. The estimated number of tracks traversed by Fe and Si ions was very similar to that of biophysical method of quantification.
Bibliography Type:	Description: (Last Updated: 08/14/2018)
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